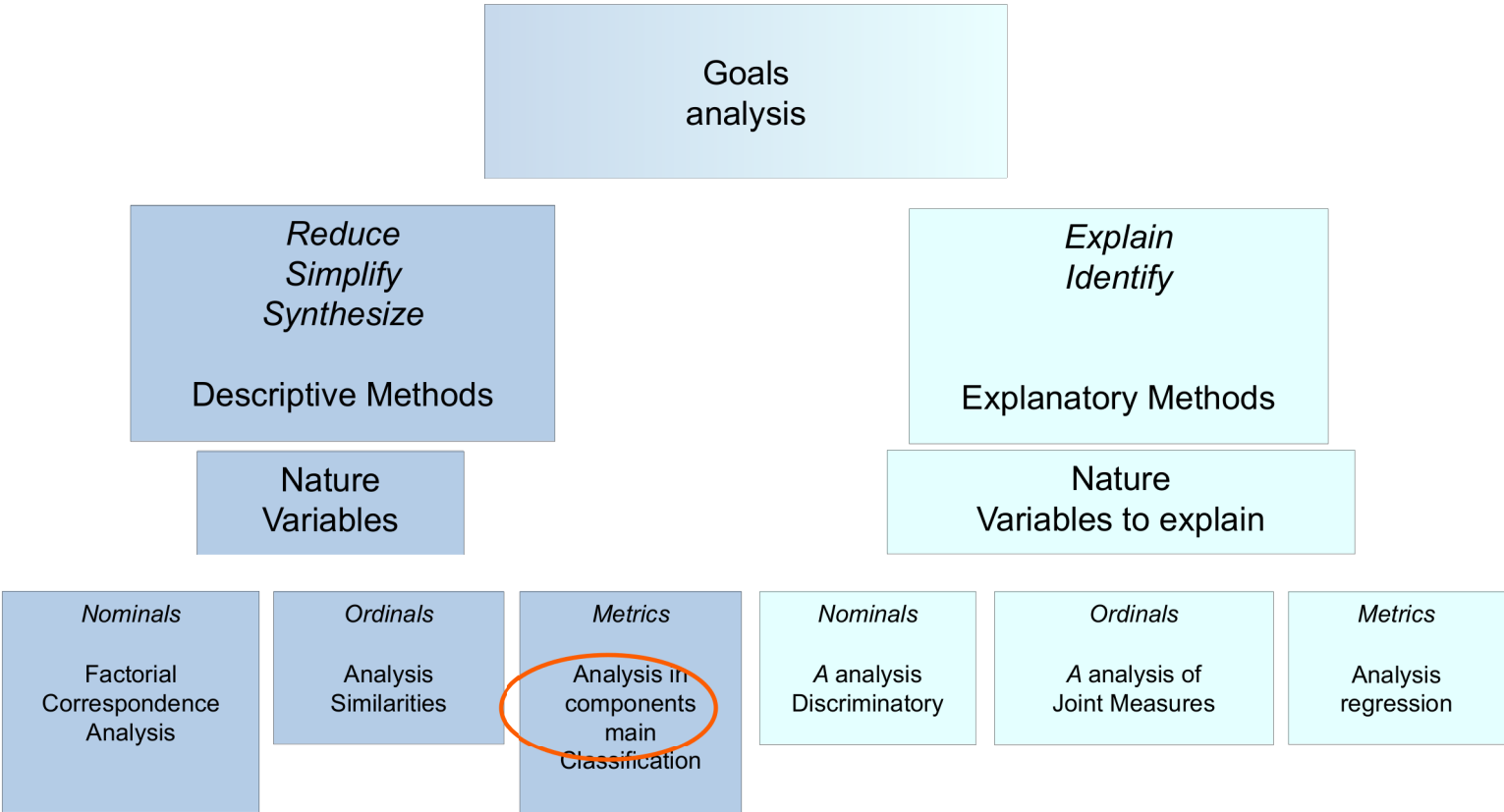


Principal Component Analysis (PCA)

Marketing Data Analysis

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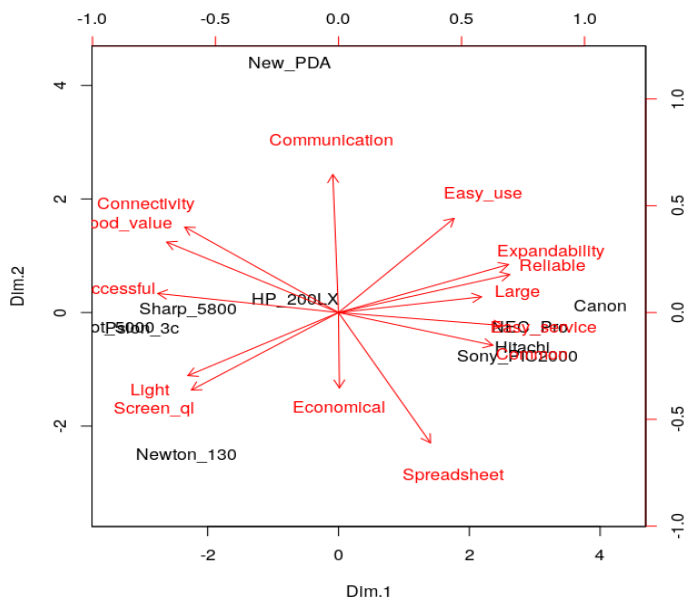
Multivariate Analysis



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Applications

- Reduce, Simplify, Synthesize
- to obtain perceptual maps when positioning a product;
- to transform a set of variables into linearly independent variables required for multiple analyses (Multicollinearity).



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Principal Component Analysis Another synthesis

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	seg
Ind1	1	5	9	2	6	10	4	7	11	3	8	12	1
Ind2	3	2	1	7	9	8	4	6	5	10	12	11	2
Ind3	2	3	1	7	9	8	4	6	5	10	12	11	2
Ind4	1	5	9	2	7	10	4	6	11	3	8	12	1
Ind5	1	3	2	7	9	8	4	6	5	10	12	11	2
Ind6	1	5	9	3	6	10	4	7	11	2	8	12	1
Ind7	2	4	9	1	6	10	3	7	11	5	8	12	1
Ind8	1	5	9	2	6	10	3	7	11	4	8	12	1
Ind9	2	5	9	3	6	10	1	7	11	4	8	12	1
Ind10	1	5	8	2	6	10	4	7	11	3	9	12	1
Ind11	1	5	9	2	6	10	3	8	11	4	7	12	1
Ind12	1	4	8	2	6	10	3	7	11	5	8	12	1

- With agglomerative hierarchical classification, synthesis occurs by associating individuals within different groups
- Ex: ...consumers in segments

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Principal Component Analysis

Another synthesis

Model	CylCap	Power	Length	Width	Mass	SpeedMax
Alfasud_TI	1350	79	393	161	870	165
Audi_100	1588	85	468	177	1110	160
Simca_1300	1294	68	424	168	1050	152
Citroen_GS	1222	59	412	161	930	151
Fiat_132	1585	98	439	164	1105	165
Lancia_Beta	1297	82	429	169	1080	160
Peugeot_504	1796	79	449	169	1160	154
Renault_16	1565	55	424	163	1010	140
Renault_30	2664	128	452	173	1320	180
Toyota_Corolla	1166	55	399	157	815	140
Alfetta_1.66	1570	109	428	162	1060	175
Princess_1800	1798	82	445	172	1160	158
Datsun_200L	1998	115	469	169	1370	160
Taunus_2000	1993	98	438	170	1080	167
Rancho	1442	80	431	166	1129	144
Mazda_9295	1769	83	440	165	1095	165
Opel_Rekord	1979	100	459	173	1120	173
Lada_1300	1294	68	404	161	955	140

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Principal Component Analysis (PCA) – Objectives

- Simplify a raw data table by going from a large number of variables to the same number of new variables but with a small number of synthetic ones
- Structure and interpret the initial data,
 - thanks to these new variables
 - and by allowing certain graphic representations (notably in the plan)

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Orthogonal projection on 2 axes

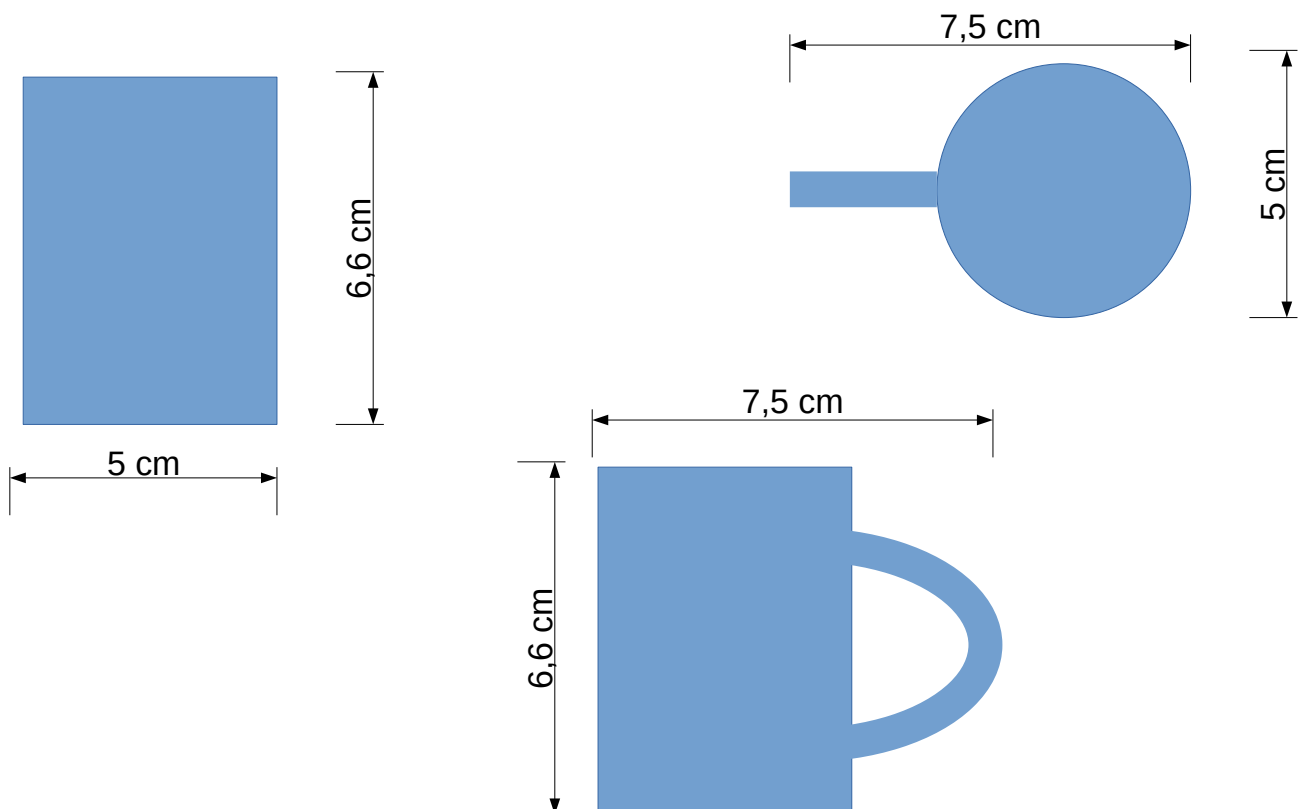
Which axes to choose?



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Orthogonal projection on 2 axes

Which axes to choose?

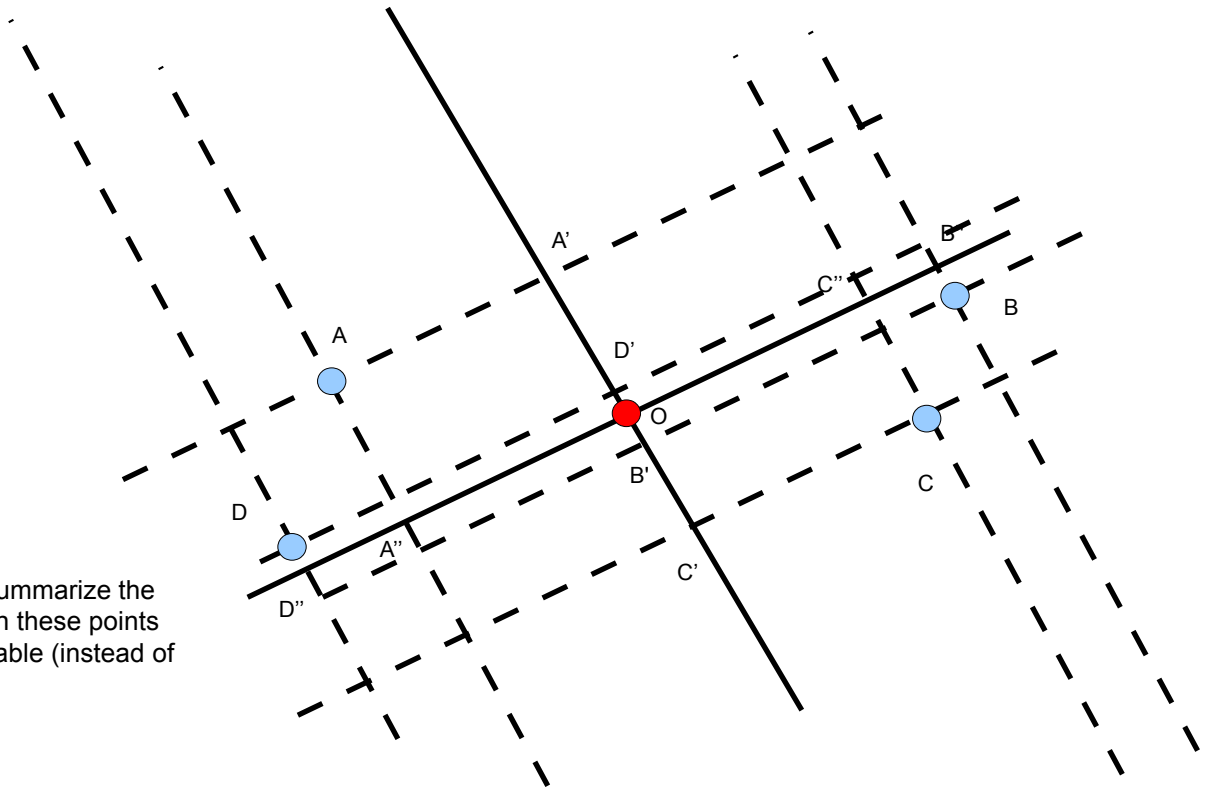


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Orthogonal projection on 1 axis

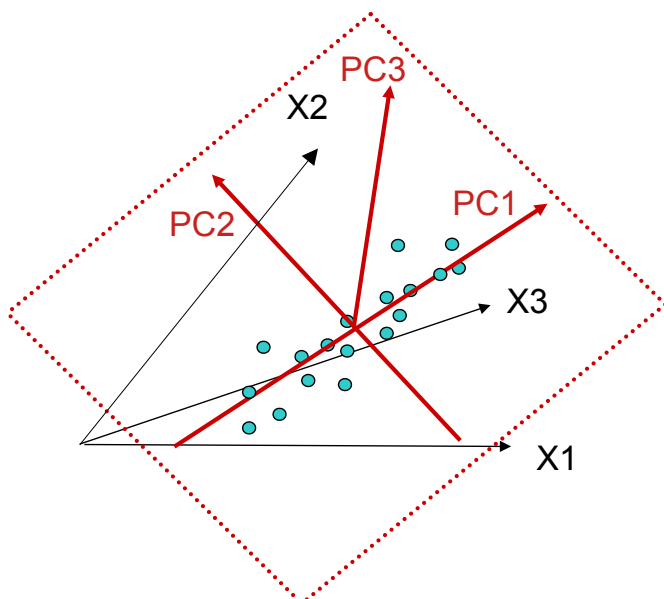
Which axis to choose?

How can we best summarize the differences between these points using only one variable (instead of 2)?



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Search for the Factorial Plan



- X_1 , X_2 and X_3 correlated
- **PC1, PC2 and PC3 linearly independent**

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Main properties of the results obtained

- The operation can therefore be seen as a **change of axes** replacing any system of axes (defined by the initial variables) by another (defined by the principal components) having two major properties
 - the axes are **orthogonal** (the **new variables are independent**);
 - the total variance is **concentrated on the first axes** (on the first new variables).
- It is this property of concentration of variance which will make it possible to **reduce the dimensionality** of the problem studied by selecting only part of the axes identified by the analysis.

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Principal Component Analysis (PCA) - Problem formulation

- Restore as best as possible the differences (Sum of the Variances) carried by the initial p variables between the objects studied by creating new variables
- A reduced number of new variables should make it possible to restore a very significant part of the “initial differences” (*Inertia*) between the objects .

Define the new variables so that they reflect the largest possible part of the Total Variance (Inertia).

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PCA results

Correlations between initial variables

```
> print(data_acp)
```

	PRICE	QUALITY	ADV	DIST
M1	112	10	132	99
M2	92	19	413	99
M3	122	1	2062	93
M4	82	5	1346	101
M5	68	15	908	98
M6	51	8	972	94
M7	120	9	1330	90
M8	119	7	1085	111
M9	66	11	337	92
M10	87	4	842	115

```
> cor(data_acp)
```

	PRICE	QUALITY	ADV	DIST
PRICE	1.0000000	-0.2999978	0.3363051	0.1086661
QUALITY	-0.2999978	1.0000000	-0.6634920	-0.2038443
ADV	0.3363051	-0.6634920	1.0000000	-0.1243516
DIST	0.1086661	-0.2038443	-0.1243516	1.0000000

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Results of the PCA

Eigenvalues

The eigenvalues in PCA serve to indicate the amount of variance explained by each principal component. How much information or variability in the original data is captured by each principal component ? The higher the eigenvalue of a component, the more variance it explains, and thus, it represents a more important aspect of the data. We select all factors for which the eigenvalue is greater than 1 (more than 1/n variance explained).

```
> # Implementation of Principal Component Analysis
> res_pca <- PCA(data_acp, scale.unit = TRUE, graph = TRUE, axes = c(1, 2))
> # Displaying eigenvalues
> print(res_pca$eig)
```

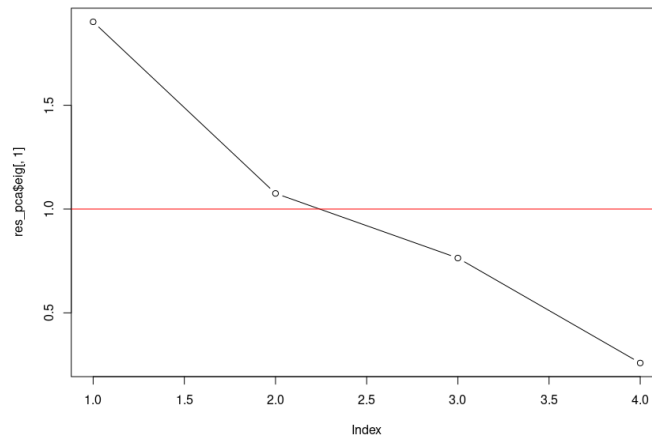
	eigenvalue	percentage of variance	cumulative percentage of variance
comp 1	1.9018475	47.54619	47.54619
comp 2	1.0754329	26.88582	74.43201
comp 3	0.7640789	19.10197	93.53398
comp 4	0.2586408	6.46602	100.00000

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Number of Axes to remember

- “Scree-Plot” analysis (or Cattell method)

```
> # Scree plot of eigenvalues  
> plot(res_pca$eig[,1], type = "b")  
> abline(h = 1, col = "red")
```



- Selection of axes for which the explained % of the total inertia is greater than $1/n$ (n : number of variables)

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PCA results Correlation table

```
> # Displaying correlations between original variables and new variables  
> print(res_pca$var$cor)
```

	Dim.1	Dim.2	Dim.3	Dim.4
PRICE	0.6263382	0.13271278	0.7668821	-0.04449414
QUALITY	-0.8707701	-0.07401738	0.3522397	0.33497461
ADV	0.8528645	-0.34806245	-0.1777035	0.34626031
DIST	0.1546916	0.96498401	-0.1425465	0.15670632

Table of correlations between the initial variables and the new variables.

This table allows you to make sense of the factors

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PCA Results

Table of Individual Coordinates

```
> # Displaying individual coordinates
> print(res_pca$ind$coord)
```

	Dim.1	Dim.2	Dim.3	Dim.4
M1	-0.6953958	0.57141721	1.1264132	-0.96043255
M2	-1.8822357	0.16248555	1.0202547	0.65036027
M3	2.7547577	-1.17330296	0.1629770	0.01945891
M4	0.7951545	-0.03092971	-0.8624267	0.10444683
M5	-1.2742060	-0.33622229	-0.3361892	0.79666075
M6	-0.6951368	-0.84842899	-1.4544972	-0.14125519
M7	0.8249648	-1.20356301	1.0737533	0.03381072
M8	1.0823284	1.50432162	0.5253548	0.30326762
M9	-1.5501154	-0.65751017	-0.3881569	-0.68325805
M10	0.6398843	2.01173276	-0.8674829	-0.12305932

Table of Values of new variables

This table provides us with the coordinates of the objects.

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PCA Results

Table of Individual Contributions

```
> # Displaying individual contributions to the principal components
> print(res_pca$ind$contrib)
```

	Dim.1	Dim.2	Dim.3	Dim.4
M1	2.542661	3.03615074	16.6057022	35.66454798
M2	18.628262	0.24549699	13.6231959	16.35351028
M3	39.901675	12.80079750	0.3476276	0.01463997
M4	3.324508	0.00889546	9.7343331	0.42178733
M5	8.536967	1.05116214	1.4792086	24.53860224
M6	2.540767	6.69341417	27.6877435	0.77145711
M7	3.578452	13.46958953	15.0893586	0.04419893
M8	6.159457	21.04253668	3.6121622	3.55594531
M9	12.634335	4.01995923	1.9718615	18.04980396
M10	2.152917	37.63199755	9.8488067	0.58550688

Allows us to diagnose which objects with exceptional characteristics led to the emergence of different factors. The sum of contributions for each axis is equal to the number of objects or 100 (depending on the software).

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PCA results

Table of cosines 2 of Individuals

```
> # Displaying squared cosines to evaluate the representation of individuals
> print(res_pca$ind$cos2)
```

	Dim.1	Dim.2	Dim.3	Dim.4
M1	0.16112031	0.1087909704	0.422748102	0.3073406189
M2	0.70390229	0.0052455839	0.206814775	0.0840373530
M3	0.84391261	0.1530914708	0.002953814	0.0000421083
M4	0.45555388	0.0006892685	0.535896765	0.0078600857
M5	0.65353465	0.0455032431	0.045494297	0.2554678051
M6	0.14473752	0.2156113181	0.633674629	0.0059765335
M7	0.20728641	0.4412022069	0.351163197	0.0003481840
M8	0.30807875	0.5951481131	0.072585373	0.0241877659
M9	0.69593894	0.1252125998	0.043637283	0.1352111761
M10	0.07837615	0.7746785778	0.144046527	0.0028987449

Indicator of the quality of representation of objects. An object is well represented in the plane if for these axes the sum of the cosines² is high enough (>0.6).

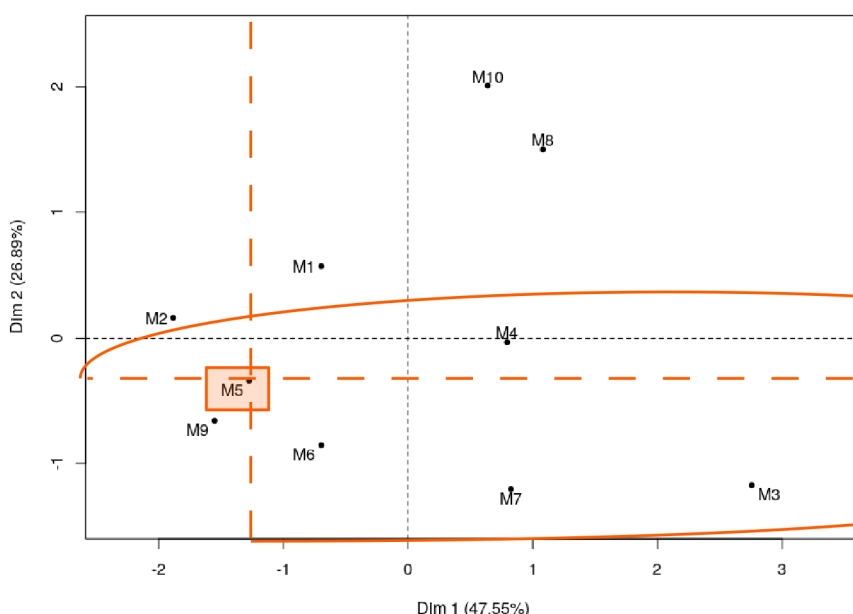
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PCA results

Graphical representations

```
> # Implementation of Principal Component Analysis
> res_pca <- PCA(data_acp, scale.unit = TRUE, graph = TRUE, axes = c(1, 2))
```

Individuals factor map (PCA)



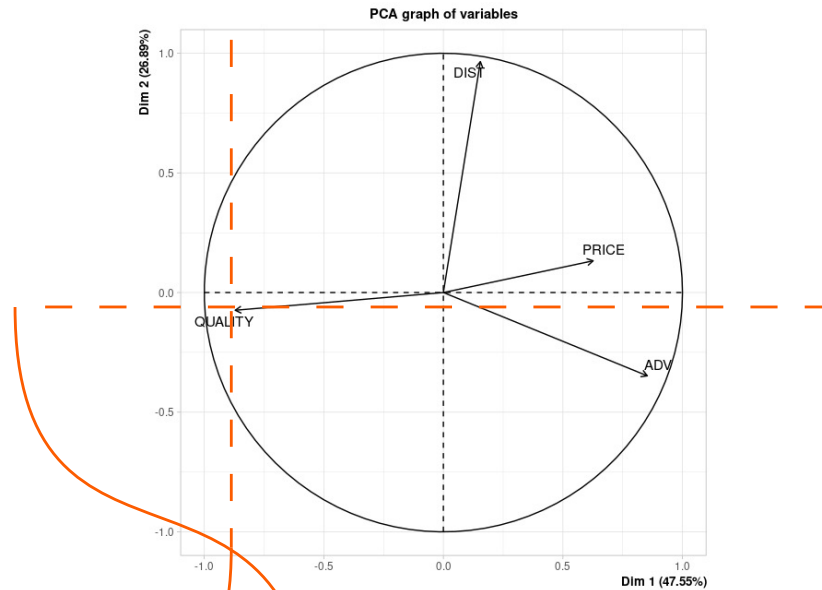
```
> # Displaying individual coord
> print(res_pca$ind$coord)
```

	Dim.1	Dim.2	
M1	-0.6953958	0.57141721	1.1
M2	-1.8822357	0.16248555	1.0
M3	2.7547577	-1.17330296	0.1
M4	0.7951545	-0.03092971	-0.8
M5	-1.2742060	-0.33622229	-0.3
M6	-0.6951368	-0.84842899	-1.4
M7	0.8249648	-1.20356301	1.0
M8	1.0823284	1.50432162	0.5
M9	-1.5501154	-0.65751017	-0.3
M10	0.6398843	2.01173276	-0.8

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PCA results

Graphical representations



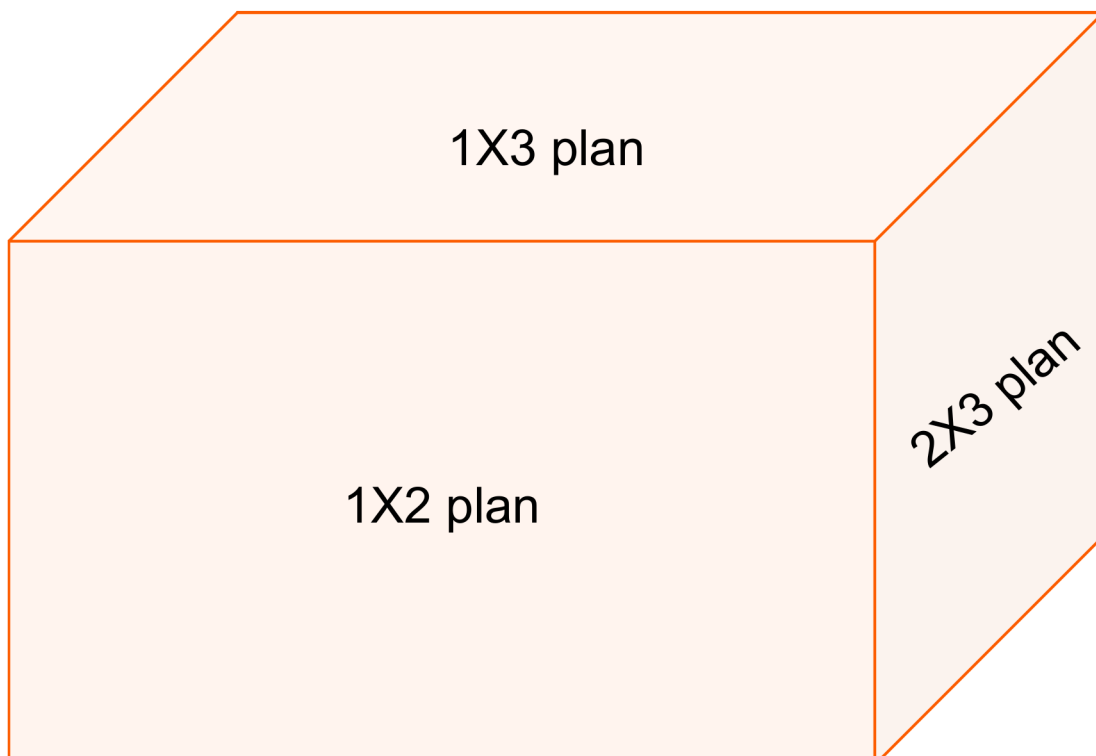
```
> # Displaying correlations between original variables and new variables
> print(res_pca$var$cor)
```

	Dim.1	Dim.2	Dim.3	Dim.4
PRICE	0.6263382	0.13271278	0.7668821	-0.04449414
QUALITY	-0.8707701	-0.07401738	0.3522397	0.33497461
ADV	0.8528645	-0.34806245	-0.1777035	0.34626031
DIST	0.1546916	0.96498401	-0.1425465	0.15670632

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Multidimensional space

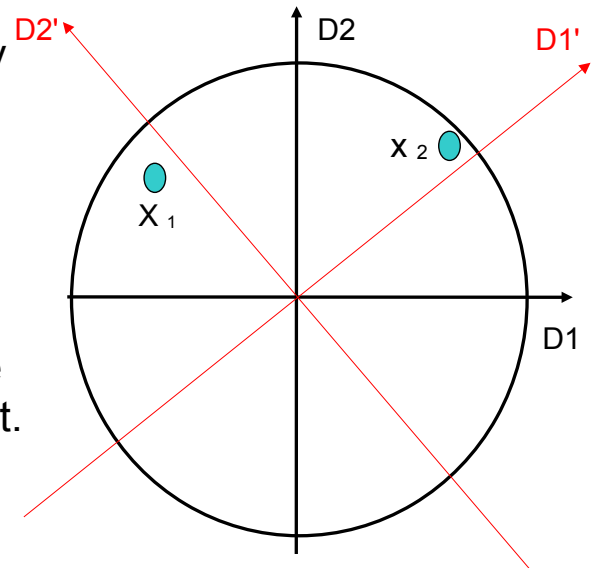
Example for 3D



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VARIMAX Rotation

- In this example, variables X_1 and X_2 are correlated with both D_1 and D_2 .
- After rotation of the axes, X_1 and X_2 are only correlated with one of the new axes.
- Interest in the rotation: Facilitate the interpretation of the axes by MAXimizing the VARiance of the correlations between the variables and the chosen axes (**VARIMAX**)
- The information restored by the vector space (here the axes D_1' and D_2') remains constant.
- There are other rotations (see specialized work)



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Example of VARIMAX Rotation

```
> print(res_pca$eig[1:2,])
```

	eigenvalue	percentage of variance	cumulative percentage of variance
comp 1	4.4208581	73.68097	73.68097
comp 2	0.8560623	14.26770	87.94867

```
> print(res_pca$var$cor[,1:2])
```

	Dim.1	Dim.2
cyl	0.8934635	0.1149061
puissance	0.8868580	0.3846891
longueur	0.8861548	-0.3810287
largeur	0.8135364	-0.4127359
masse	0.9051875	-0.2245325
vmax	0.7547104	0.5735194

If the synthesis capacity of the first 2 axes remains the same (cumulative 87.9%), it is better distributed over the 2 axes. We went from 73% for axis 1 and 14% for axis 2 to 48% for axis 1 and 39% for axis 2

```
> sol_varimax<-varimax(res_pca$var$cor[,1:2])
> # Capacité de synthèse après rotation
> print(sol_varimax$loadings)
```

Loadings:

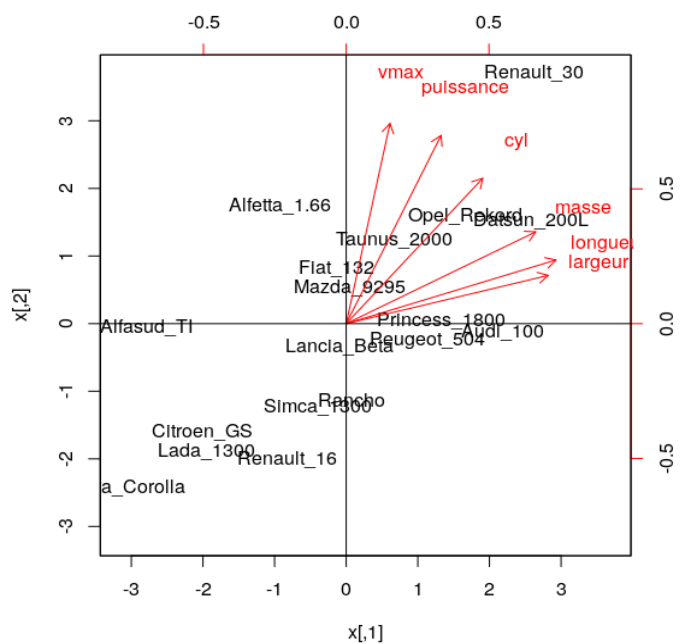
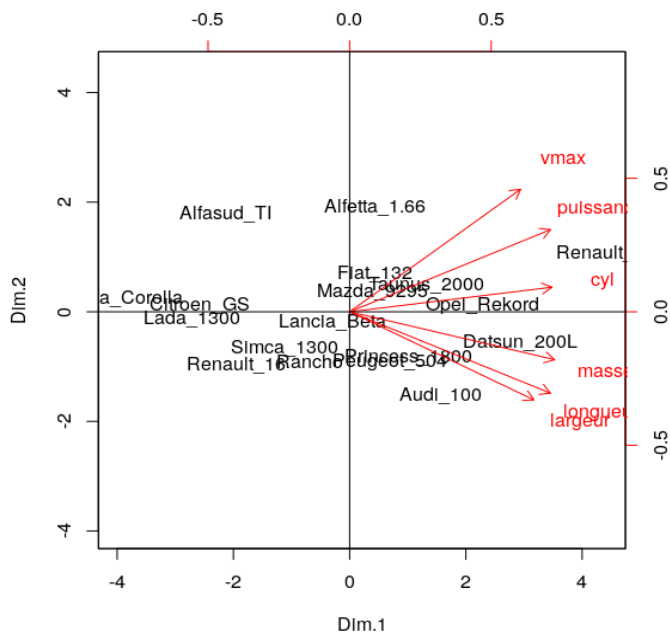
	Dim.1	Dim.2
cyl	0.598	0.674
puissance	0.415	0.873
longueur	0.918	0.295
largeur	0.884	0.224
masse	0.830	0.426
vmax	0.192	0.928

The variance of the correlations between the initial variables and the axes (new variables) increased. We went from a situation where all the variances are close (89%, 88%,....) to a situation where some are very high and others low (59%,41%,91%...Variance increased , it is even maximum).

	Dim.1	Dim.2
SS loadings	2.88	2.397
Proportion Var	0.48	0.399
Cumulative Var	0.48	0.879

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Example of VARIMAX Rotation



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Steps in Using PCA for Scale Construction

- **Data Collection:** Gather data from a sample using a set of items (questions) designed to measure a psychological construct. For example, if you are measuring anxiety, you might have items related to feelings of nervousness, worry, and physical symptoms.
- **Correlation Matrix:** Calculate the correlation matrix for the items. This matrix shows how each item correlates with every other item, which is essential for PCA.
- **PCA Execution:** Perform PCA on the correlation matrix. This involves:
 - Extracting the principal components (PCs), which are linear combinations of the original items.
 - Determining the number of components to retain, often based on the eigenvalues (e.g., retaining components with eigenvalues greater than 1).
- **Component Interpretation:** Analyze the correlations and loadings of each item on the retained components. Correlations and Loadings indicate how much each item contributes to a component. Items that contribute highly on the same component are likely measuring the same underlying construct.
- **Scale Development:** Based on the PCA results, you can develop a scale by selecting items that contribute significantly on the desired components. This helps ensure that the scale is both reliable and valid.
- **Validation:** After developing the scale, it's important to validate it through additional testing.

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Items from the “Trust” measurement scales

CREDIBILITY

- The products of this brand provide me with security. (CREDI1)
- I trust the quality of the products from this brand. (CREDI3)
- Buying products from this brand is a guarantee. (CREDI4)

INTEGRITY

- This brand is sincere towards consumers. (INTE2)
- This brand is honest with its customers. (INTE4)
- This brand shows interest in its customers. (INTE5)

GOODWILL

- I believe that this brand updates its products to take into account advancements in research. (GOODW1)
- I think this brand continuously seeks to improve its responses to consumer needs. (GOODW2)